Inventory Image Tagging, Phase 2

A small CNN, an ablation study and a hybrid. Same data and the same fair test as Phase 1.

87.61%

Phase 1 best
(Pixels + Random Forest)

91.98%

Small CNN
(mean of 3 seeds)

92.49%

Hybrid: CNN features
+ Random Forest

Phase 1 mixed up shirts, pullovers and coats

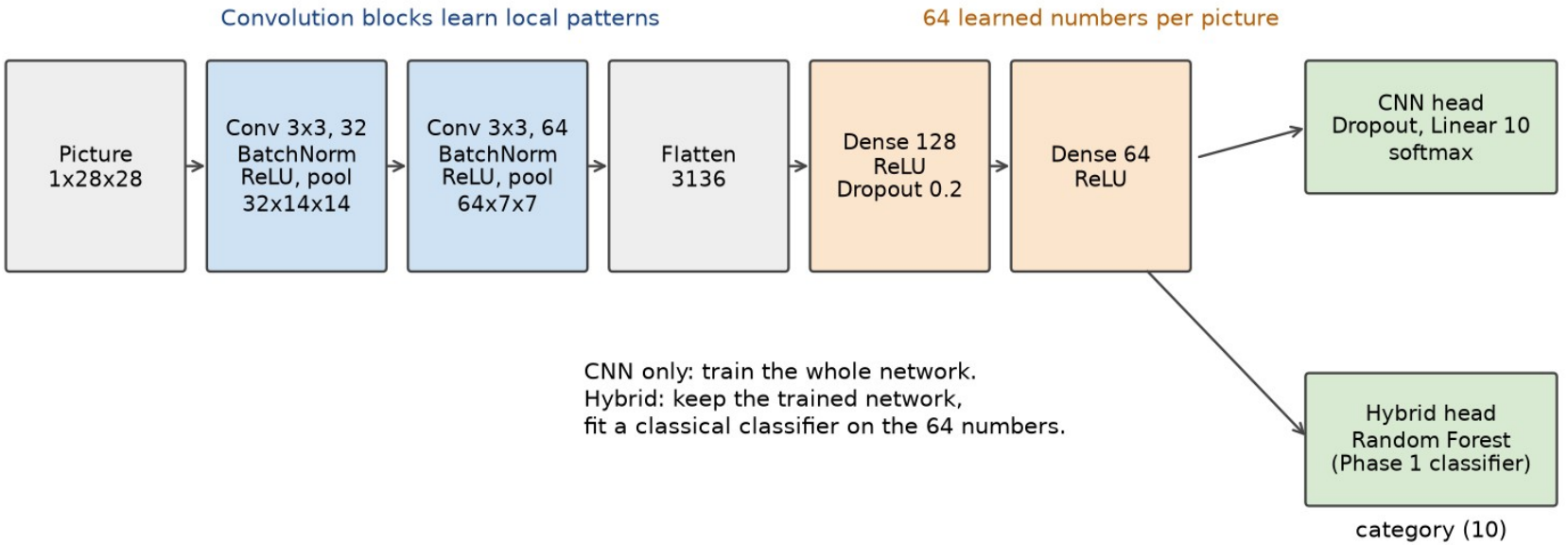
Phase 1: fixed description

- Pixels or HOG edges, chosen before the model sees any data
- Vector models have no built-in local filters
- Only 59.2% of shirts found

Phase 2: learned description

- Small filters learn edges and shapes from the pictures themselves
- Same filter is used everywhere, so few weights are needed
- Question: does that fix the similar categories?

Two conv blocks, then 64 learned numbers

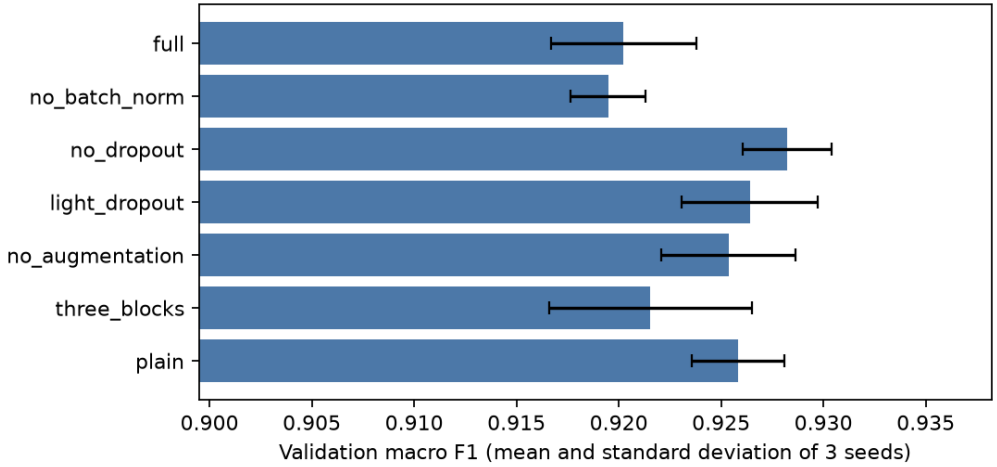


429,450
trainable weights

28 → 14 → 7
size after each pooling

64
numbers the hybrid reuses

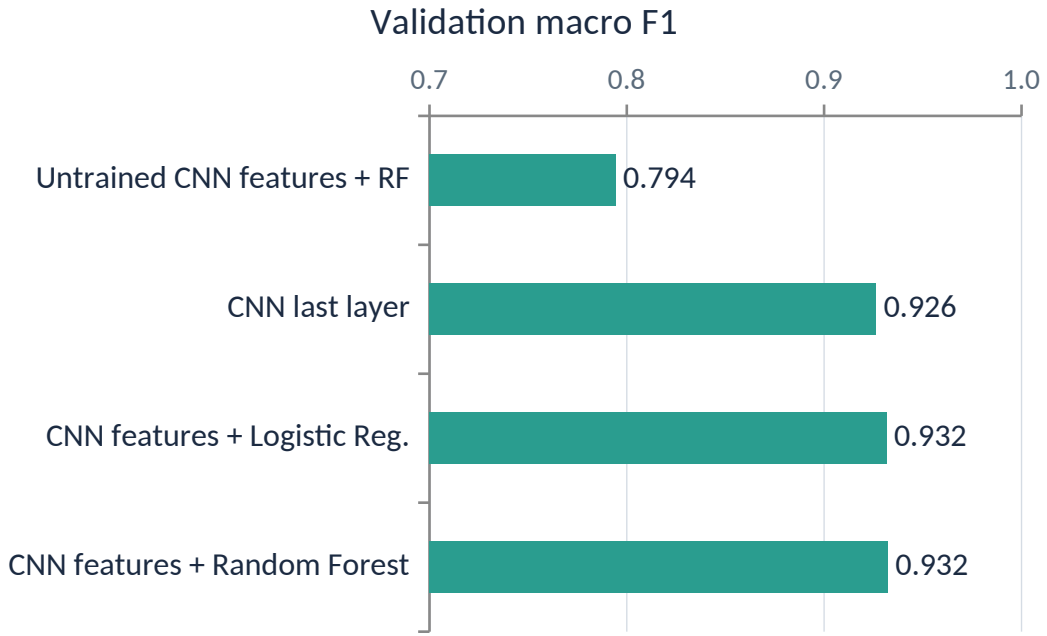
Ablation: most choices sit within the noise



Each version trained with 3 seeds. Bars show the mean and the spread.

- All seven within 0.9 points
- Dropout 0.4 hurt; 0.2 is about the same as none
- Batch norm, augmentation and a third block gave no clear gain
- Final CNN: best version with dropout, as the brief asks

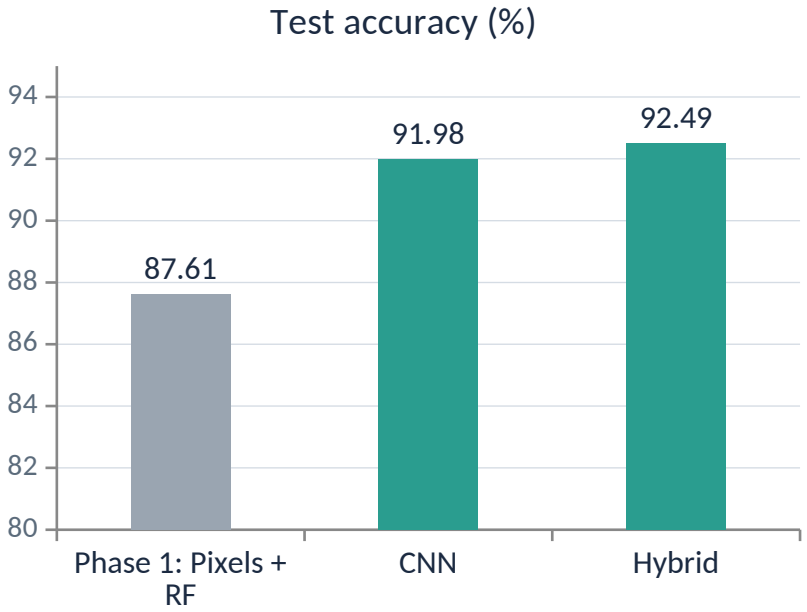
Hybrid: a classifier on the CNN features



The chart axis starts at 0.7.

- The CNN is trained first. A Phase 1 classifier is then fitted on its 64 numbers.
- Untrained features score far lower, so training makes them useful
- Every classifier gains about half a point: the gain is small but real

The CNN beats Phase 1 by about 4 points



The chart axis starts at 80%.

Recall	Phase 1	CNN	Hybrid
Shirt	59.2%	73.6%	75.7%
Pullover	80.2%	88.7%	89.6%
Coat	82.2%	87.0%	88.2%
T-shirt/top	85.2%	88.9%	88.5%

Paired bootstrap, 95% interval: CNN over Phase 1 +4.10 points [+3.52, +4.67]; hybrid over CNN +0.60 [+0.28, +0.94]. Pullover → Coat mistakes fell from 116 to 33.

Limits and how to rerun

Limits

- Fashion-MNIST is small and familiar, not a store catalogue
- One split, 3 seeds, small search
- Apple GPU training is not exactly repeatable
- Not in the live dashboard

Rerun it

Install PyTorch:

```
bash scripts/setup_mac.sh --phase2
```

Notebook (no download):

```
notebooks/phase2.ipynb
```

Retrain everything (about 45 min):

```
bash scripts/build_phase2.sh
```